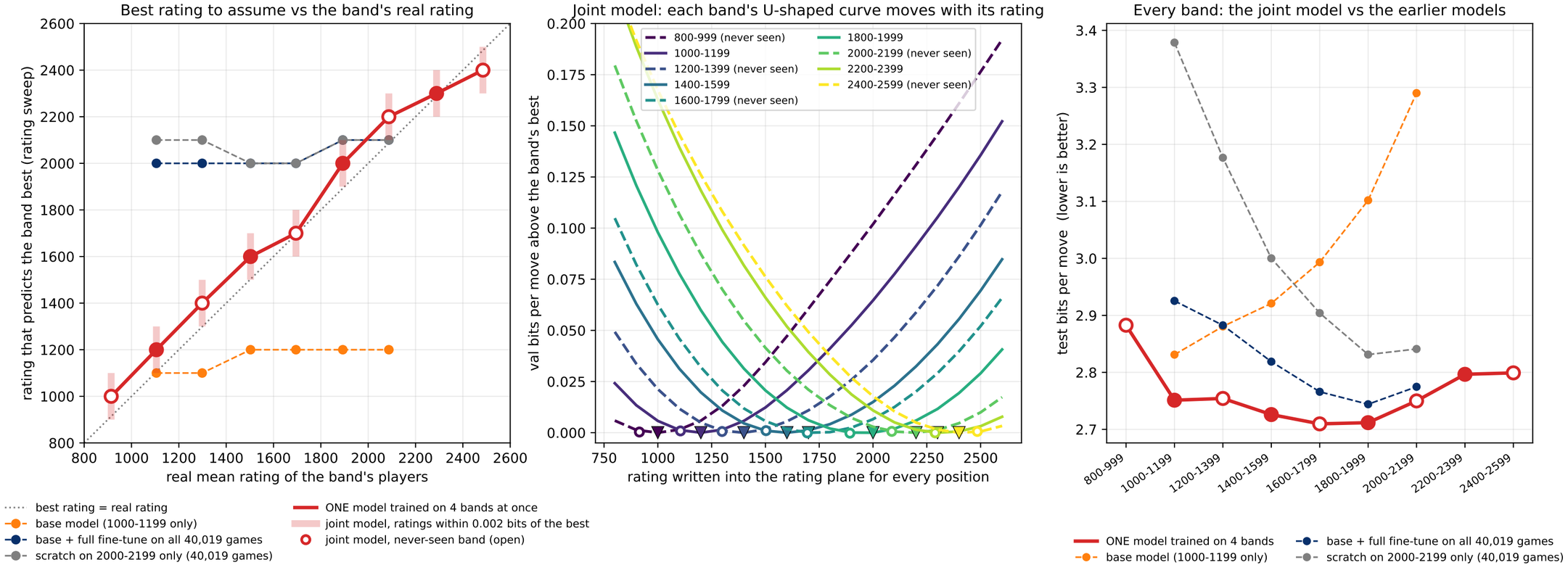
Trained on 4 rating bands at once, the model's best rating to assume rises with each band's real rating



Left: for each band, the rating that gives the lowest val bits when written into the rating plane for every position, against the band's real mean rating. The earlier models (trained on one band, or one band after another) prefer one fixed rating for every band; the joint model's best rating rises with the real rating, including the never-seen bands (open circles). The bottom of each curve is flat (shaded bars: ratings within 0.002 bits of the best), so the 100-point best rating is coarse; the rating plane also stands in for time control and game length, which change with rating. Middle: the joint model's sweep; ▼ = best rating, ○ = real ratings. Right: test bits (shard 000 test rows); open circles = bands the joint model never trained on. This comparison is not controlled: the joint model ran 1.9× the steps of the base model, trained on the bands next to each never-seen band, and some players of the never-seen bands also play in the training bands. Trained bands: 30,000 games each; 2 passes; one seed.